Answer **one** question in this section in the spaces provided.

7 **(a)** A glass tube, open at both ends, has fine dust sprinkled along its length. A sound source is placed near one open end of the tube, as shown in Fig. 7.1.

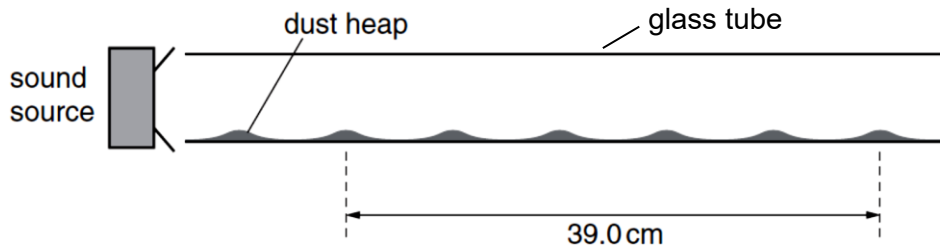

Fig. 7.1

(i) On Fig. 7.1, sketch the form of the stationary wave set up in the tube and mark, with the letter A, the position(s) of any displacement antinode(s) of the stationary wave. [2]

(ii) The distance between six heaps, as shown in Fig. 7.1, is 39.0 cm.

Show that the wavelength of the stationary wave is 15.6 cm.

[1]

(iii) The frequency at which these heaps are formed is 2.14 kHz.

Calculate the speed of sound in the tube.

speed = ... m s^{-1} [2]

(iv) State the fundamental frequency at which a stationary wave can be set up in the tube.

fundamental frequency = ... Hz [3]

(v) State and explain the change, if any, to the dust heaps if the right open end of the tube is blocked with a lid, keeping the frequency at 2.14 kHz.

……..

……. [2]

(b) Fig. 7.2 shows an arrangement used to determine the speed of microwaves in the laboratory.

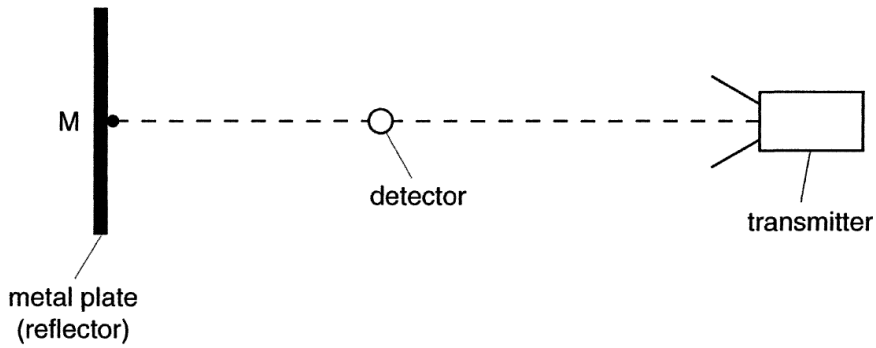

Fig. 7.2

A stationary wave is formed between the metal plate and the transmitter. As the detector is moved along a line between the transmitter and the point M, it registers a series of high intensity signals. The positions of these high intensity signals are marked by crosses on an axis as shown in in Fig. 7.3.

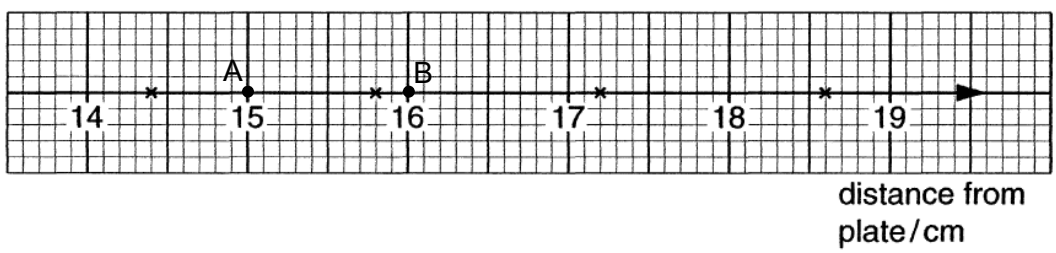

Fig. 7.3

(i) State the phase difference between points A and B.

phase difference = …… rad [1]

(ii) The transmitter emits microwaves of frequency 10 GHz.

Use Fig. 7.3 to deduce an experimental approximation for the speed of the microwaves.

speed of microwaves = …… m s^{-1} [2]

(iii) The detector is now fixed at a position of high intensity.

If the metal plate is moved away from transmitter at a speed of 1.0 cm s^{-1} while the detector is kept stationary, determine the frequency of the high intensity signals registered on the detector.

frequency = ... Hz [2]

The metal plate is now removed from the arrangement and a sheet of hardboard is inserted between the detector and the transmitter without either of these being moved as shown in Fig. 7.4. The hardboard is a soft boundary that partially reflects microwaves.

When the hardboard is inserted, the intensity of the wave drops to $\frac{1}{16}$th of its original value.

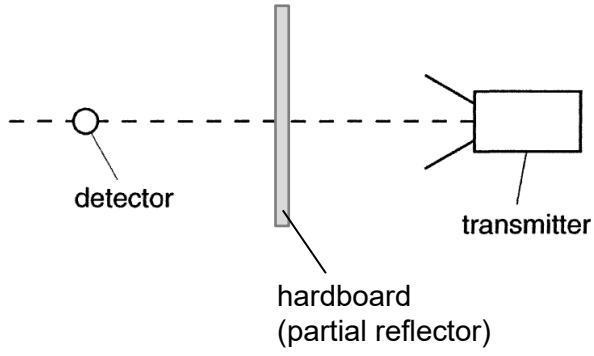

Fig. 7.4

(iv) Calculate the ratio of $\frac{\text{amplitude of detected wave after hardboard insertion}}{\text{amplitude of detected wave before hardboard insertion}}$.

ratio = ... [2]

The metal plate is now placed back in original position as shown in Fig. 7.5.

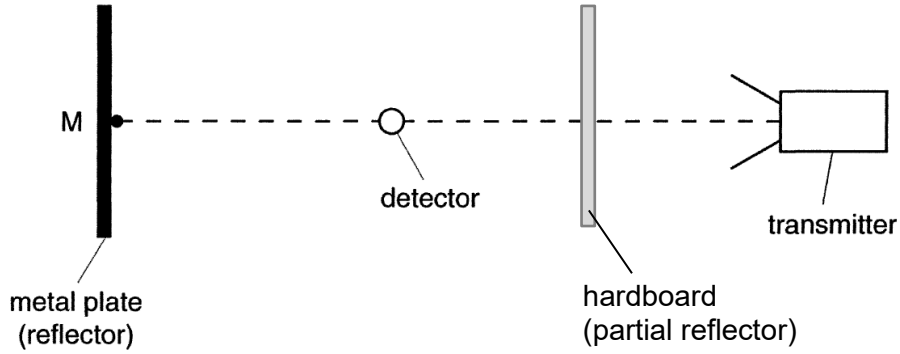

Fig. 7.5

(v) Fig. 7.6 shows the variation of intensity with position as the detector is moved between the transmitter and M at a particular frequency *before* the hardboard is inserted.

Sketch on Fig. 7.6 the variation of intensity with position as the detector is moved between the hardboard and M *after* the hardboard is inserted.

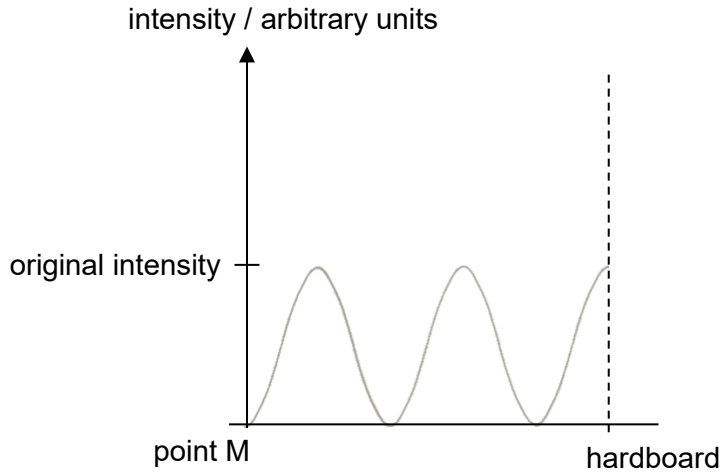

Fig. 7.6

[3]